

Student:

Jalen Salgado

Email:

Time on Task:

1 hour, 51 minutes

Progress:

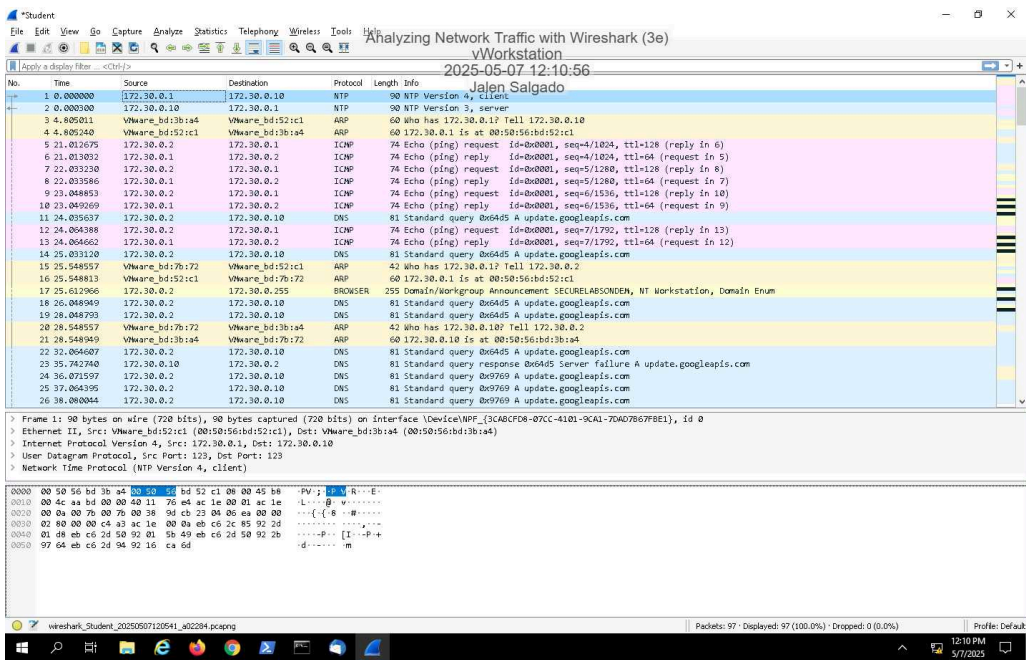
100%

Report Generated: Saturday, July 26, 2025 at 2:11 PM

Section 1: Hands-On Demonstration

Part 1: Explore Wireshark

13. Make a screen capture showing the fields related to time.



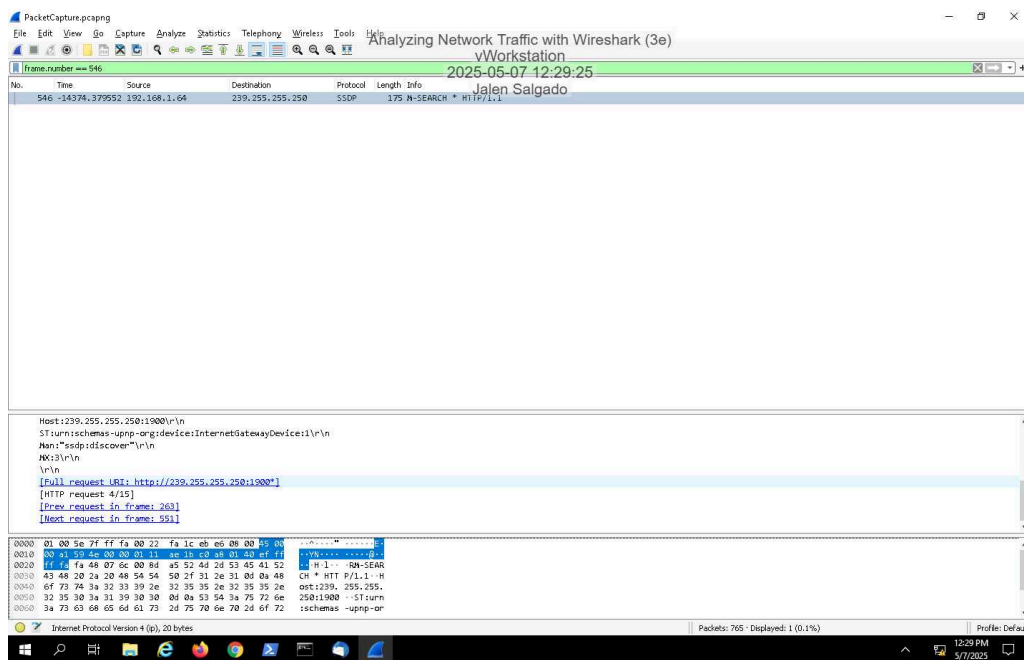
Part 2: Analyze Wireshark Capture Information

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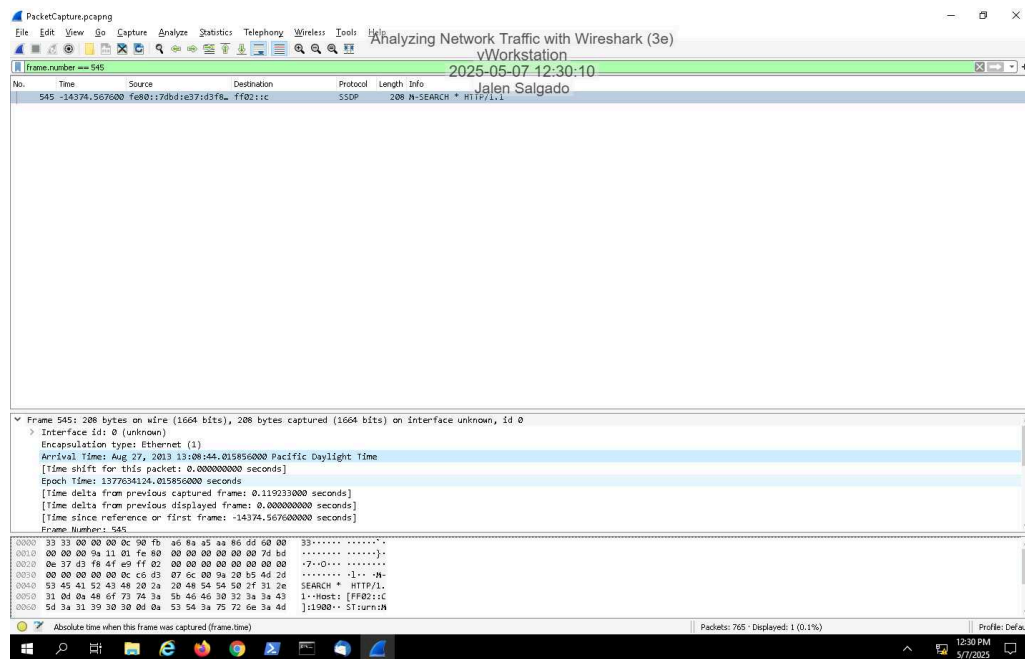
13. Record the source IP address in Packet 546.

192.168.1.64

19. Make a screen capture showing the related frame numbers for Packet 546.



- Make a screen capture showing the **complete hexadecimal representation for the source and destination Media Access Control (MAC) addresses in Packet 545.**



- Record the **IEEE-assigned manufacturer's unique ID in Packet 545.**

90:fb:a6

- Record the **MAC address used for multicast in Packet 545.**

33:33:00:00:00:0c

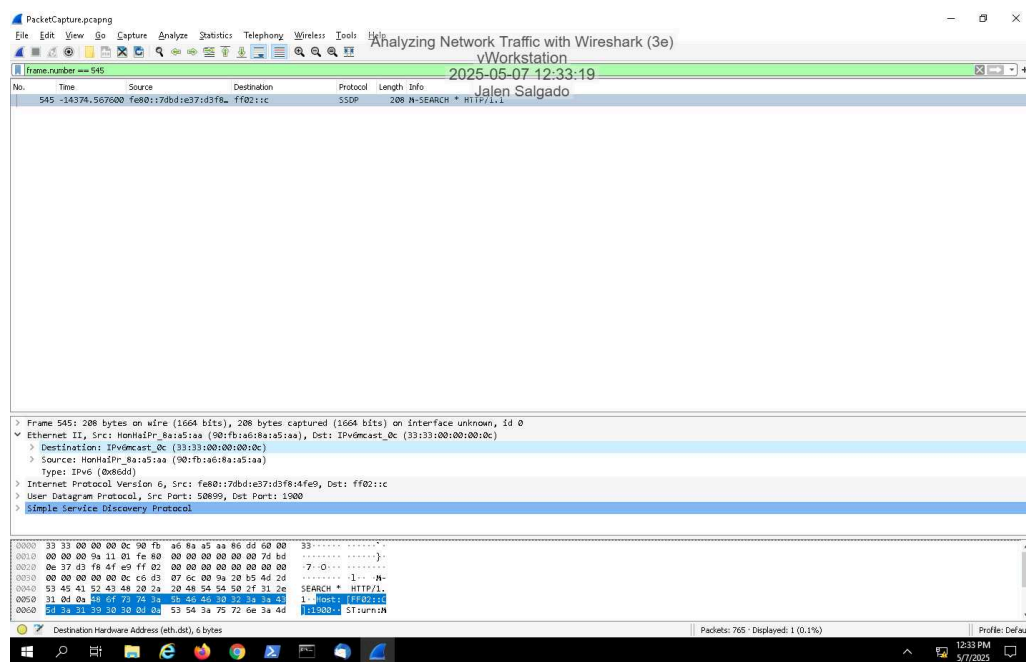
- Record the **version of the Internet Protocol being used in Packet 545.**

IPv6

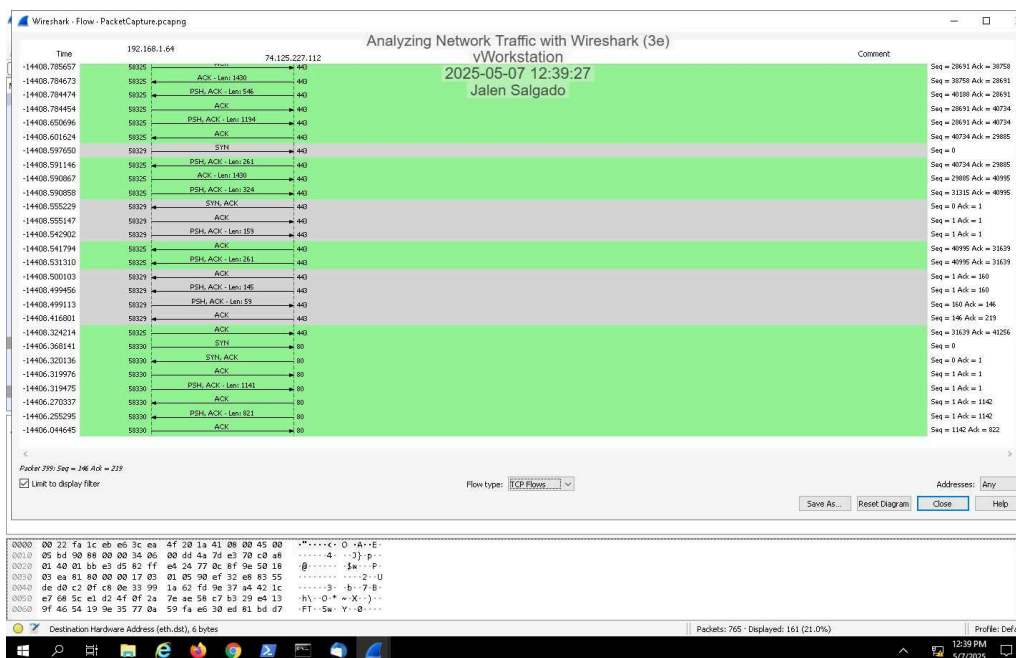
- Record the source IP address in Packet 545.

fe80::7dbd:e37d:3f8:4fe9

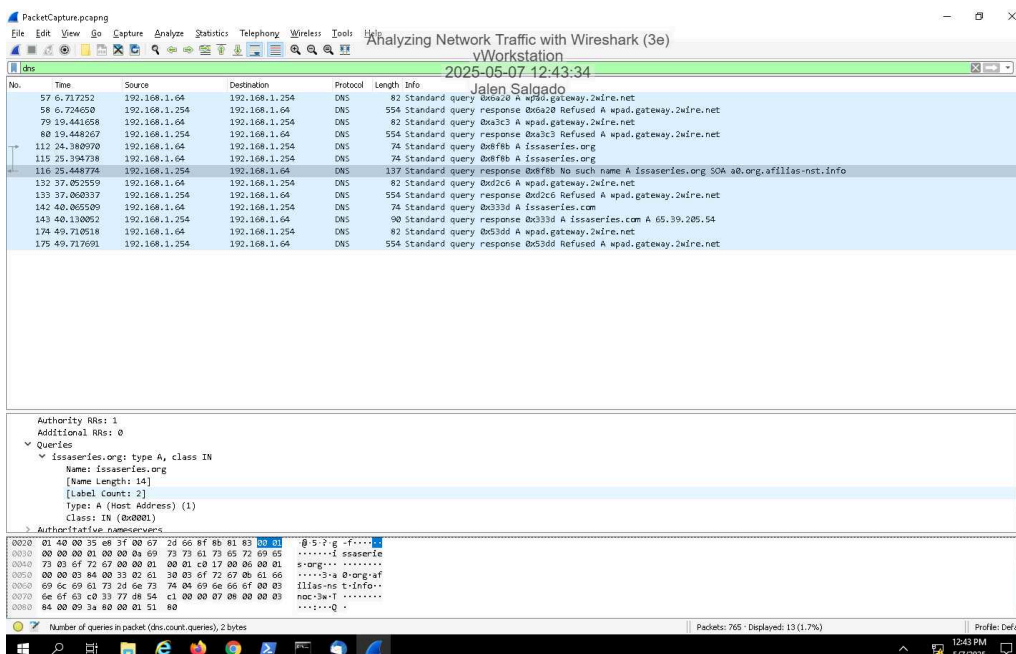
- Make a screen capture showing the related frame numbers for Packet 545.



36. Make a screen capture showing the **time** (found in the Time column on the left) that each step of the handshake occurred.



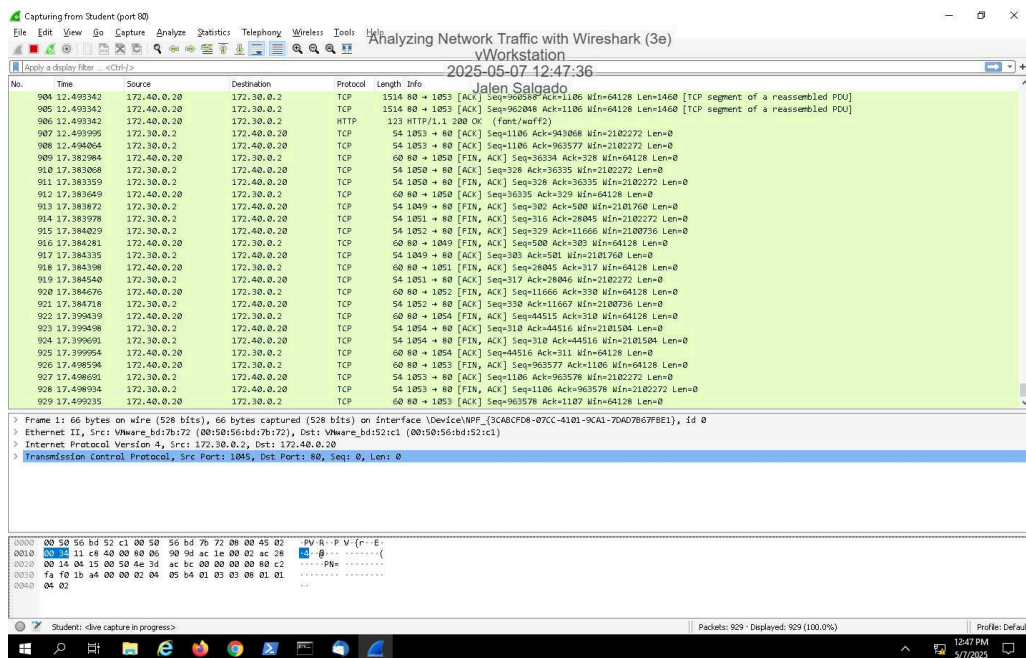
45. Make a screen capture showing the response to the issaseries.org query.



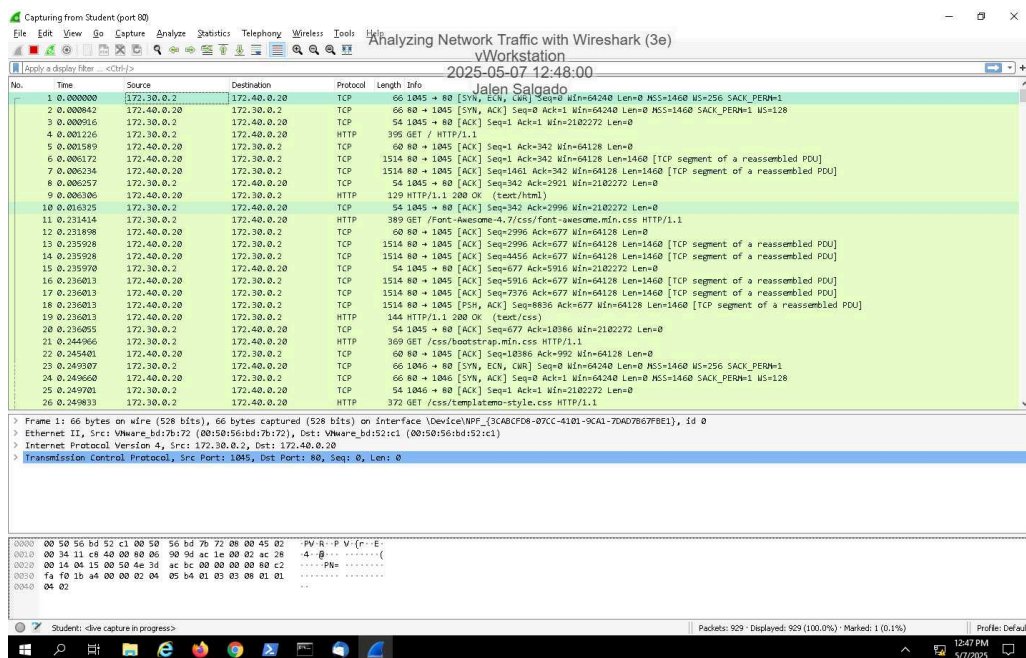
Section 2: Applied Learning

Part 1: Explore Wireshark

12. Make a screen capture showing the http traffic.



15. Make a screen capture showing the fields related to time.



Part 2: Analyze Wireshark Capture Information

- Record the number of bytes captured and the bytes on the wire.

237 on wire and 237 captured

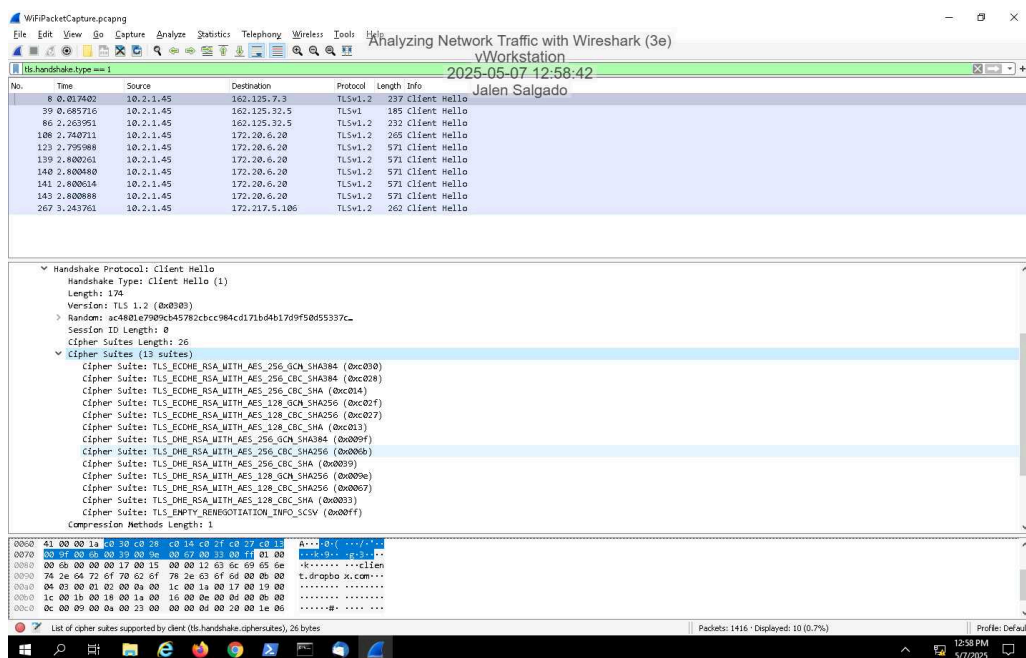
- Record the manufacturer of the destination device.

ASUSTekC_a3:ae:f8

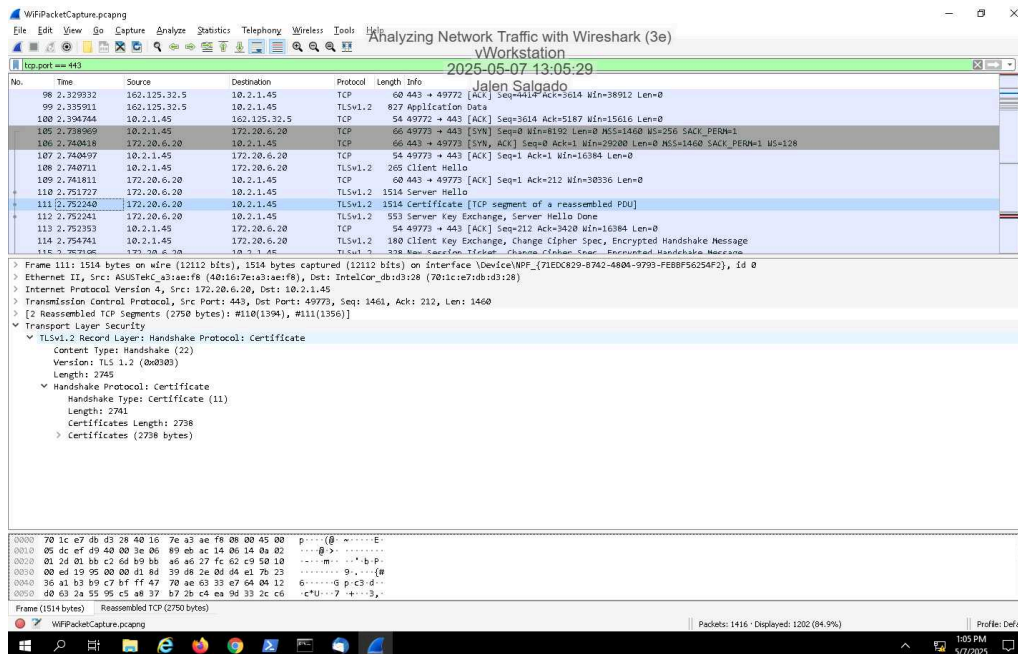
- Record the source IP address.

Version 4 SRC 10.2.1.45 Dst: 162.125.7.3

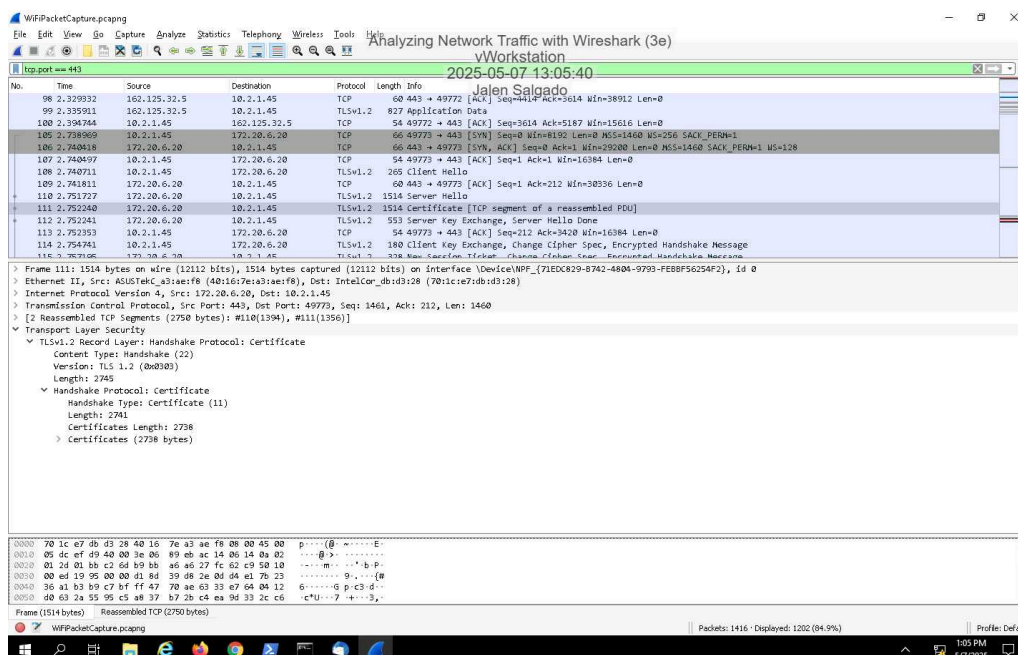
- Make a screen capture showing the entire list of cipher suites.



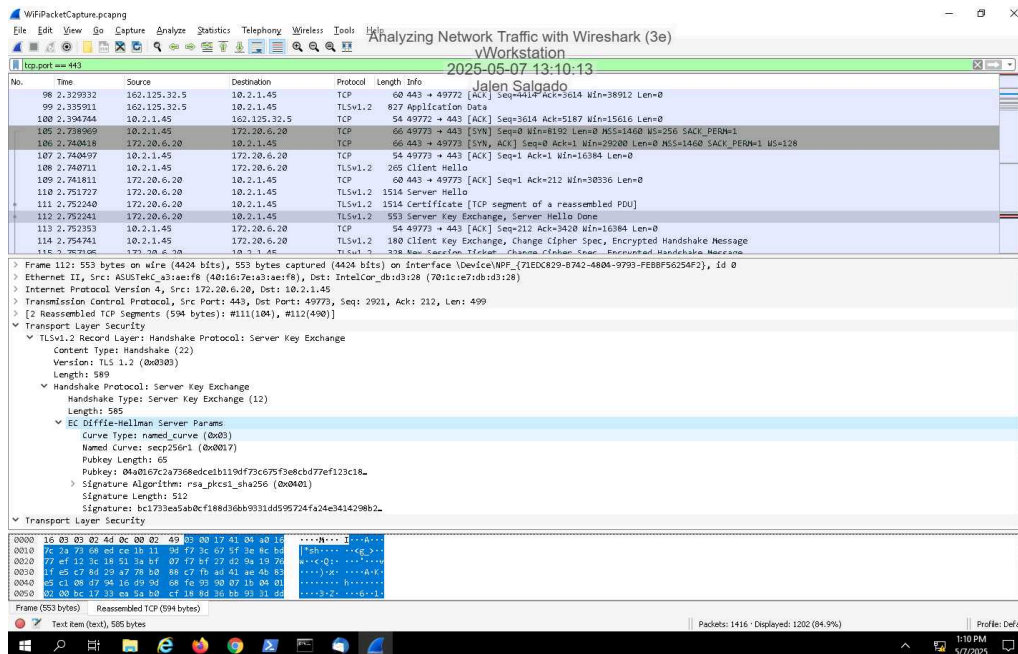
21. Make a screen capture showing the issuer of the certificate.



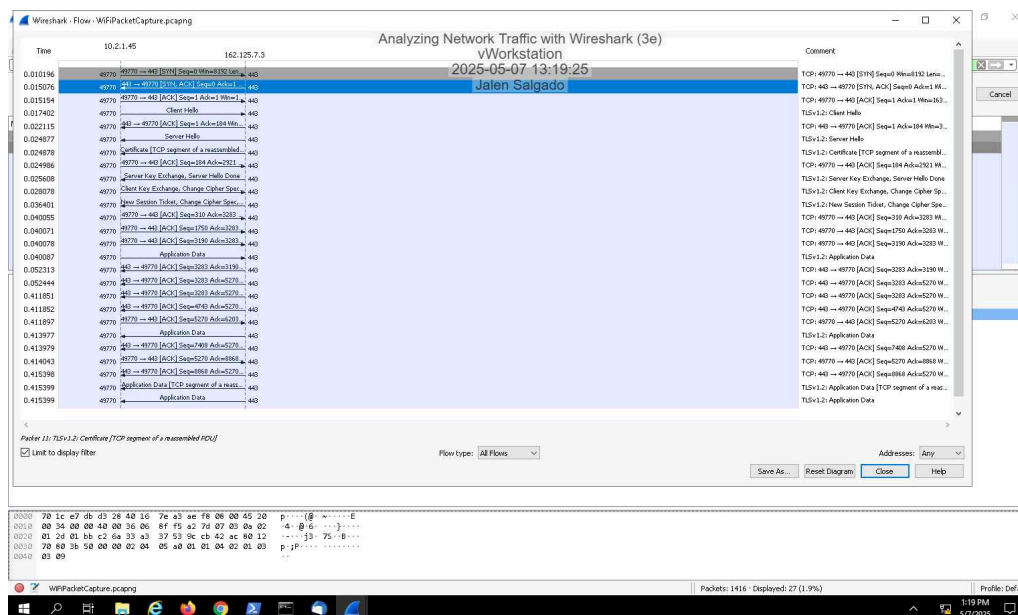
24. Make a screen capture showing the details of the certificate.



27. Make a screen capture showing the public key and signature hash for the certificate.

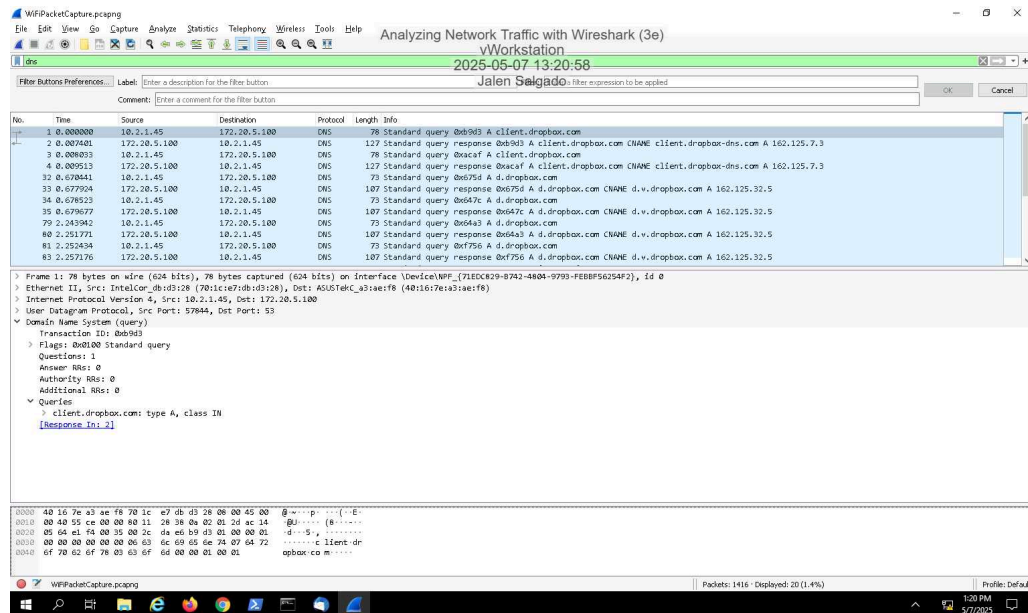


35. Make a screen capture showing the first three-way TCP handshake in the Flow Graph.



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40. **Make a screen capture** showing the **query posed in this packet**.



Section 3: Challenge and Analysis

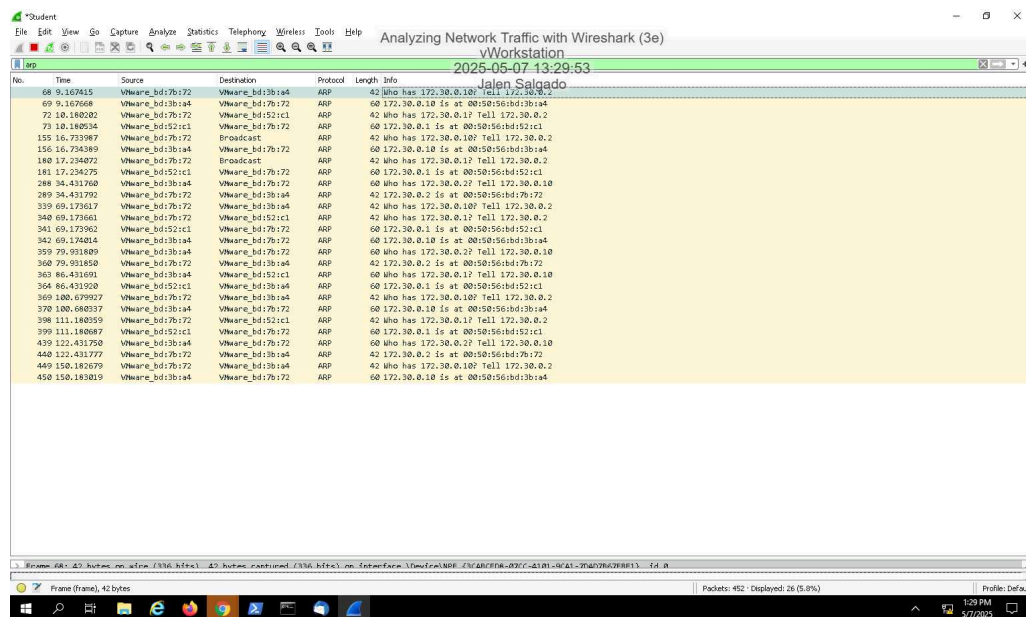
Part 1: Research Common Network Traffic

Identify at least five common protocols and their associated TCP/UDP port numbers, then **explain** their purpose and relevant features (for example, known security vulnerabilities, etc.).

HTTP: It delivers unencrypted web content and is used for browsing websites the dont use encryption its also vulneraabler to eavesdropping and MITM attacks. **HTTPS** on port 443 is the main one. It secures the connection using TLS so no one can snoop on what you're doing. It's solid, but if the site's using an outdated version of TLS, it can still get hit by things like downgrade attacks. **DNS** on port 53 is what turns the website name into an IP address so your browser knows where to go. It usually runs over UDP, and the problem is it's not encrypted by default, which makes it easy to spoof unless it's using DNS over HTTPS. Then there's just **TCP**, which handles the reliable delivery of all this traffic. It keeps things in order, but it can be abused with attacks like SYN floods. **TLS** is also a big piece here. It handles the actual encryption behind HTTPS. It's strong now, but older versions had some serious holes like Heartbleed. These protocols basically work together every time you load up a website.

Part 2: Capture and Filter Traffic Using Wireshark

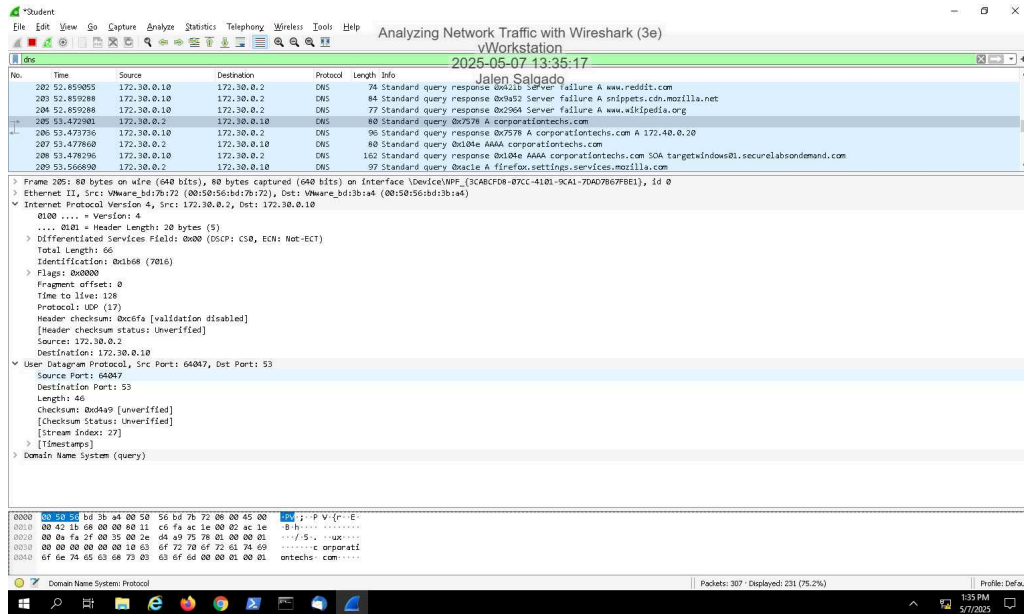
Make a screen capture showing the **MAC address resolved by ARP** for the DNS server.



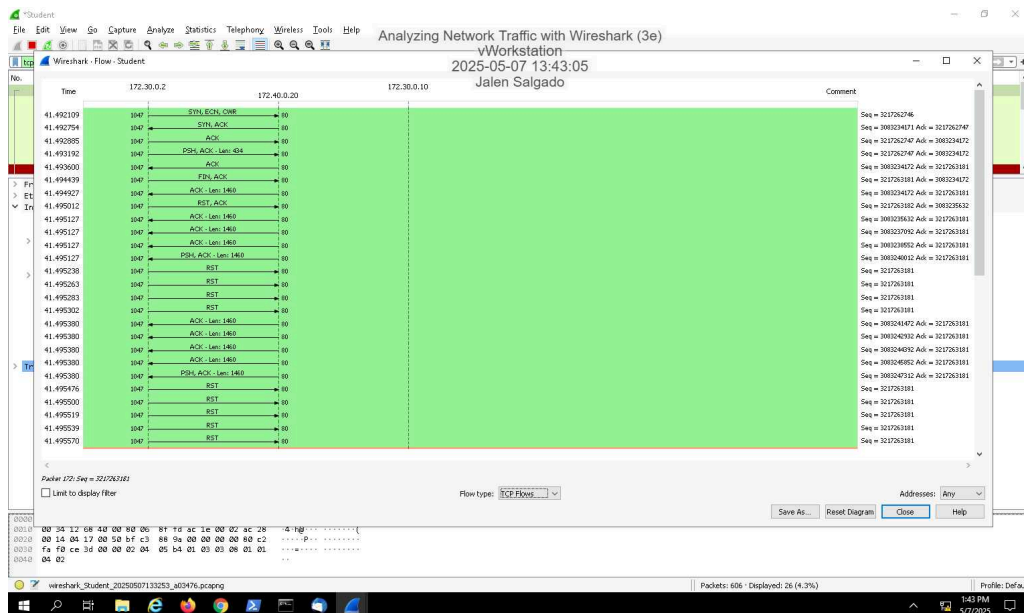
Analyzing Network Traffic with Wireshark (3e)

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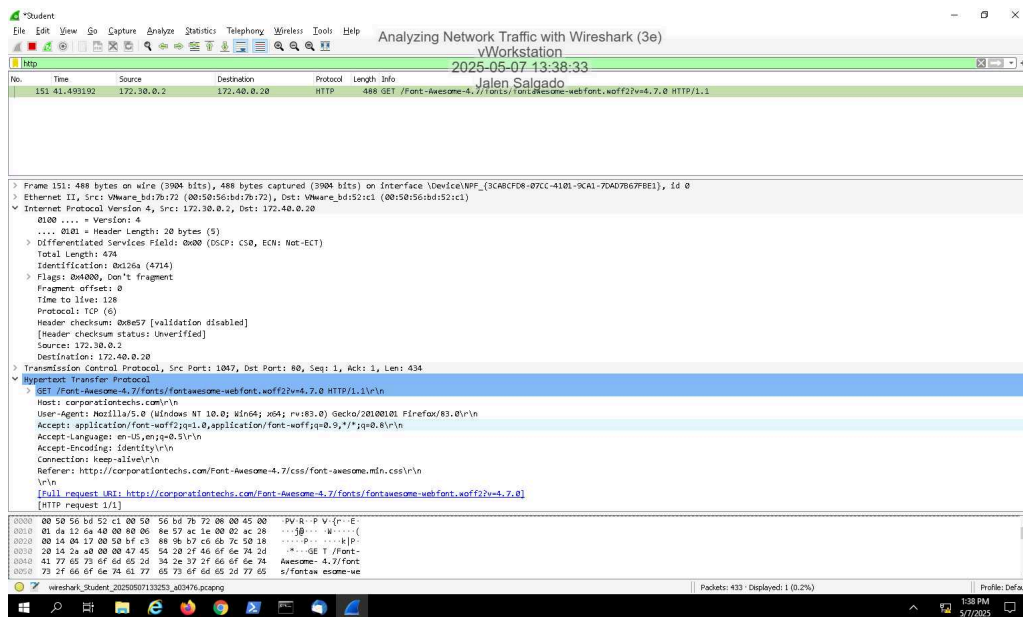
Make a screen capture showing the destination IP address and port number of the DNS server.



Make a screen capture of the three-way handshake that took place between the client PC and the web server.

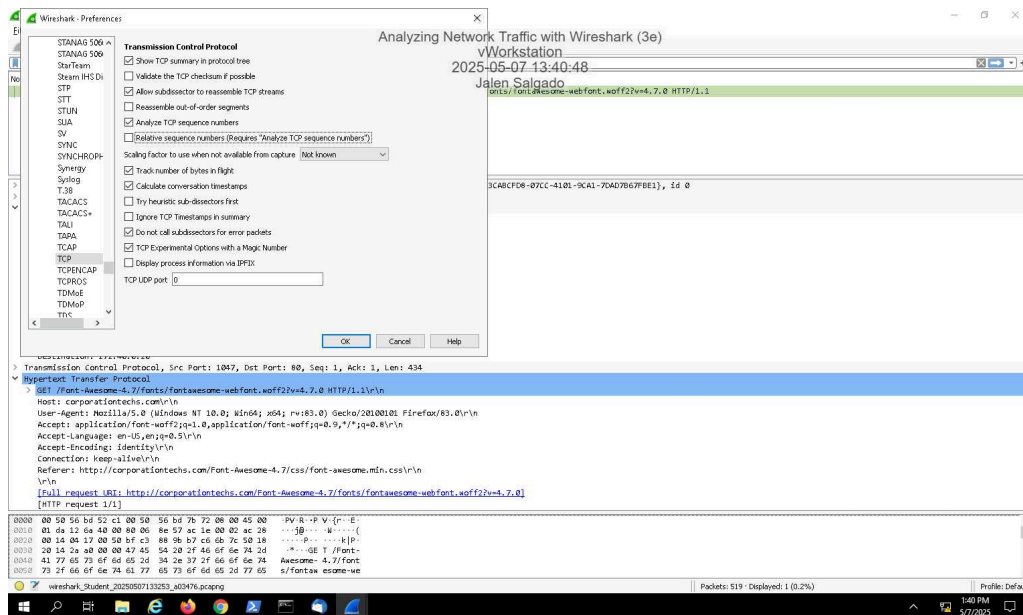


Make a screen capture of the actual HTTP traffic that was delivered from the corporationtechs.com web server.



Part 3: Analyze Capture Files

Make a screen capture showing the updated Wireshark TCP preferences for relative sequence numbers.



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Make a screen capture showing the flow graph displaying the sequence and acknowledgement values recorded during the three-way handshake.

